

SF Planning Commission Minutes: Pipeline Status and Preliminary Results

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1 Purpose

This directory documents the empirical backbone of the project: turning twenty-eight years of San Francisco Planning Commission meeting minutes (1998–2026) into an *item-level* dataset of discretionary land-use decisions. One row is one request heard by the Commission — a conditional use, a discretionary review, a variance, a rezoning — with its case number, address, zoning district, staff recommendation, disposition, roll-call vote, and hearing date.

That dataset is what makes the “market for housing regulation” measurable. It lets us ask who obtains discretionary approval, how often staff recommendations are overridden, how the approval rate moves with the political composition of the Commission, and how any of this responds to state preemption. The theory side lives in `output/political_economic_housing_model/`; the extension of the same pipeline to the rest of the Bay Area is documented in the recon directories.

2 What is in this directory

The directory is **one folder per memo**. Each holds `<name>.tex`, its compiled `.pdf`, and its own `figures/` and `tables/`, so a memo and everything it depends on move together; every figure and table is written by a script, and each memo’s last section names the script that regenerates it. `README.md` in this directory is the index. The spec and reference documents the pipeline code links to by path live in `notes/`.

Folder	What it answers
<code>memo/</code>	This memo: pipeline status and what is still open. Edited in place.
<code>meeting_level_info/</code>	How each item is matched to the meeting it was heard at, the meeting-level fields, and their accuracy against the gold set.
<code>extraction_method_comparison/</code>	Regex against Claude, field by field, on a frozen train/test split, with accuracy and over-extraction reported separately.
<code>discretionary_review_patterns/</code>	What the 16,199-item corpus contains: composition of the docket, dispositions, delay, commissioners, geography, Planning Code citations.
<code>permit_linkage/</code>	How far the entitlement can be followed to a real building permit in DBI’s records, and what the match buys.

Folder	What it answers
<code>conditions_of_approval/</code>	Whether the <i>content</i> of the Commission’s conditions can be recovered, given that <code>conditions_imposed</code> is only a yes/no flag.
<code>predicting_delay/</code>	How much elapsed hearing time is forecastable from what a developer can observe at the first hearing — in sample and out.
<code>notes/</code>	<code>labeling_rules.md</code> (the coding manual), <code>data_infrastructure.md</code> (schema and flow), <code>minutes_data_availability.md</code> , <code>processing_review.md</code> , <code>schema_enrichment_recommendation.md</code> , <code>hand_label_review_guide.md</code> , <code>help_string_audit.md</code> .

3 The pipeline

Scrape → **parse** → **date** → **label** → **extract**. Code lives in `code/commission_minutes_processing/`; the corpus lives outside the repo on Dropbox and is reached through `paths.py`.

Parsing splits each source document into blocks, one per agenda item, delimited by case headers, numbered agenda items, section headers, adjournment markers, and draft-minutes-adoption appendages. Two document eras are handled: scraped HTML (1998–2014) and PDF (2015–present).

Dating is a separate stage, and deliberately so. A block never states its own hearing date — the meeting header above it does — so the date is a structural question about which meeting a block falls under, not a textual one about the block. It is answered by `assign_meeting_dates.py` and validated against hand-marked meeting boundaries collected in `date_boundary_app/`. See the companion memo, `meeting_level_info.tex`.

Meeting-level labelling sits above the items. Time of day, whether the sitting was regular, special, joint or closed, the room, who sat and who staffed it are properties of the hearing, shared by every item heard at it, so they are recorded once per meeting and joined on the date rather than repeated on 23,000 rows. `meeting_headers.py` cuts the header window around each known meeting start and pre-fills twelve such fields for confirmation in the boundary app.

Labeling is done in a local Flask app against the shared schema, with a machine pre-fill the human corrects. *Extraction* then either fine-tunes a model on those labels or few-shot prompts one, and `learning_curve.py` plots field accuracy against label count so labeling stops at the empirical plateau rather than a guess.

3.1 The schema is the single source of truth

`extraction_common.py` holds one list, `SCHEMA`, from which the labeling form, the model prompt, the training builder’s required keys, and the evaluation metric are all generated. Editing that list propagates everywhere. It currently carries **28 fields** in six groups: Identity, Location, Zoning & scale, Process, Politics, Conditions.

Four fields have been dropped as the schema settled, each for a stated reason: `jurisdiction` (constant), `supervisory_district` (recoverable from the address), `item` (agenda number, no

analytic signal), and — as of 30 August 2026 — `meeting_date`, which is a property of the meeting rather than the item and is now supplied by the dating stage.

Two design decisions in the schema are worth stating because they carry the empirics. First, `action` records the disposition *family* (`approved`, `disapproved`, `continued`, `took_dr_and_approved`, ...), with “with conditions” and “as modified” split out into two orthogonal flags. This removed a three-way overlap in the old vocabulary and lets conditions attach to non-approval outcomes. Second, the staff recommendation is stored twice: verbatim (`preliminary_recommendation`) and bucketed to the same disposition families (`preliminary_recommendation_category`). The gap between that category and `action` is the staff-override signal the project studies, and storing both makes “was the staff recommendation followed?” a direct enum comparison rather than string wrangling.

4 Preliminary results

4.1 Corpus

Quantity	Count
Parsed items, 1998–2026	25,927
HTML era (1998–2014)	17,080 across 724 pages
Modern era (2015–2026)	8,847
of which carry a case number	16,269
Meetings identified, HTML era	818
Meetings identified, whole corpus	1,219 across 1,125 documents
Pages holding more than one meeting	41 (max 5)

4.2 Dating

The dating stage re-dates every HTML-era item from its position in the source page. Distinct hearing dates per year moved from 12–17 to 35–47 for 1998–2000, in line with the Commission’s roughly weekly sitting; later years were already close to correct. Against the hand-marked gold standard — now 116 documents, scored on HAND marks only, since a later validation round pre-marked its documents by machine and those must not be scored as gold — boundary recall is **1.000** and precision **0.992**, positional recall **1.000** and precision **0.984**, and **item date accuracy is 1.000 on 1,752 scored items**. All modern PDFs agree with their file names. Re-derive with `python date_boundary_app/app.py --score`; details and caveats are in the companion memo.

4.3 Meetings

134 meetings are hand-confirmed, and the extraction now runs over the whole corpus rather than the sample: **1,219 meetings, 1998–2026**, written to `meetings_all.csv`. Against the hand-confirmed set the ten meeting-level fields agree on **93.9%** of 938 values (`extract_all_meetings.py --score`); the out-of-sample generalisation estimate from the last frozen round is **96.0%**. The two differ because the first covers every round under a stricter comparison and is unadjudicated; the meeting-level memo keeps them apart. The sample is already informative about the panel’s composition: 90 regular sittings, 7 special, and 5 joint — with the Redevelopment Agency Commission (1998), Building Inspection (2007, 2018), Public Health (2011) and Historic Preservation (2023).

Two joint sittings are headed simply “Special Meeting” and are identifiable as joint only from the two bodies named above the date.

4.4 Labeling

Status	Items	Meaning
<code>done</code>	289	confirmed gold
<code>flagged</code>	89	a specific issue to recheck
<code>not_an_item</code>	9,658	agenda scaffolding, not a land-use decision
<code>todo</code>	15,891	unlabeled

The queue is ordered rare-class-first and interleaved by year, so labeling top-down fills scarce request types and dispositions evenly across the panel rather than exhausting 1998 first. A fixed random sample per year is flagged separately so the gold set is representative of the whole period rather than of whichever years were labeled earliest.

5 Status and open questions

- **Settled.** The schema (v2); the parser’s block boundaries; the dating stage and its validation; the labeling workflow and coding manual; the extraction configuration (Haiku 4.5, six era-matched examples, structured outputs with evidence spans) and its corpus-wide run.
- **In progress.** Item-level hand-labeling; 289 items are confirmed gold and the frozen split is 155 train / 139 test. The unified review queue holds the remaining work. The meeting-level extraction is complete over the whole corpus (1,219 meetings) with 134 hand-confirmed.
- **Open.** The adjudication queue still holds 143 unjudged gold-vs-model disagreements, and they are built against a superseded zero-shot run rather than the live configuration — so the share of measured “model error” that is really gold error is known only for the batch already judged (54% model, 43% gold, 3% both). The learning curve has not been run at full size, so the number of labels required for a local model is still an estimate. Six commissioners’ roll-call entries still merge two surnames where the minutes omit a separator (‘Melgar Moore’, ‘Moore Sugaya’); splitting them needs a roster the project does not yet hold.
- **Decision pending.** Whether to extend meeting-level hand-confirmation beyond the 134 already done. The pre-fill is good enough that the cost is confirmation rather than transcription, but the analytic payoff of the meeting-level fields is not yet established.
- **Known non-transfer.** The heuristic extractor is tuned to SF minutes and does not transfer to other jurisdictions; extending the pipeline requires re-testing extraction on real documents with human review, not unattended.